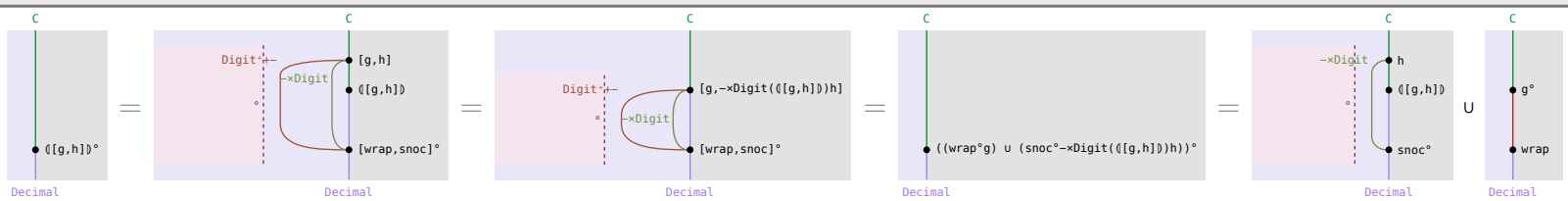
Companion to Bird & de Moor, Algebra of Programming

§6. Recursive Programs

§6.1. Digits of a number

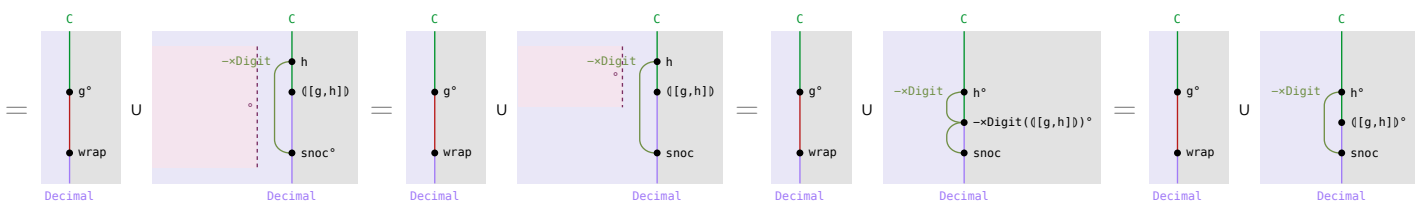
`⦇[g,h]⦈° = (g° wrap) ∪ (h° −×Digit(⦇[g,h]⦈°) snoc)` (6.1a)

the converse of a fold over decimals undoes the algebra `[g,h]` and then either `wrap`, or the fold's converse on the front and `snoc`



(1) (2) (3) (4) (5)

1	`⦇[g,h]⦈°`	`⦇φ⦈ = [wrap,snoc]° Digit⁺+−(−×Digit(⦇φ⦈)) φ`
2	`= ([wrap,snoc]° Digit⁺+−(−×Digit(⦇[g,h]⦈))) [g,h])°`	`(R + S) [P,Q] = [R P,S Q]`
3	`= ([wrap,snoc]° [g,−×Digit(⦇[g,h]⦈) h])°`	`[U,V]° [R,S] = (U° R) ∪ (V° S)`
4	`= ((wrap° g) ∪ (snoc° −×Digit(⦇[g,h]⦈) h))°`	`(R ∪ S)° = S° ∪ R°`
5	`= (snoc° −×Digit(⦇[g,h]⦈) h)° ∪ (wrap° g)°`	`R ∪ S = S ∪ R`



(6) (7) (8) (9)

6	`= (wrap° g)° ∪ (snoc° −×Digit(⦇[g,h]⦈) h)°`	`(R S)° = S° R°`
7	`= (g° wrap) ∪ ((−×Digit(⦇[g,h]⦈) h)° snoc)`	`(R S)° = S° R°`
8	`= (g° wrap) ∪ (h° −×Digit(⦇[g,h]⦈)° snoc)`	`−×Digit(R)° = −×Digit(R°)`
9	`= (g° wrap) ∪ (h° −×Digit(⦇[g,h]⦈°) snoc)`	

`val° = (embed° wrap) ∪ (op° −×Digit(val°) snoc)`

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest

(6.1b)

1	`op°(m,p) ⇔ π₁(p) = m/10 ∧ π₂(p) = m % 10` `op°` splits a number into its quotient and remainder by 10
2	`(∃p. 0 < π₁(p) ∧ op°(m,p)) ⇔ 10 ≤ m` `op°` gives a pair with a nonzero first component exactly at the numbers with two or more digits
3	`(∃d. embed°(m,d)) ⇔ 0 < m ∧ m < 10` `embed°` gives a digit exactly at the one-digit numbers

§6.2. Least fixed points

(6.2a)

`Monotonic(φ) ≜ ∀X. ∀Y. X ⊑ Y ⇒ φ(X) ⊑ φ(Y)` — φ maps the hom-set `A↛B` into itself with `X⊑Y⇒φ(X)⊑φ(Y)`.

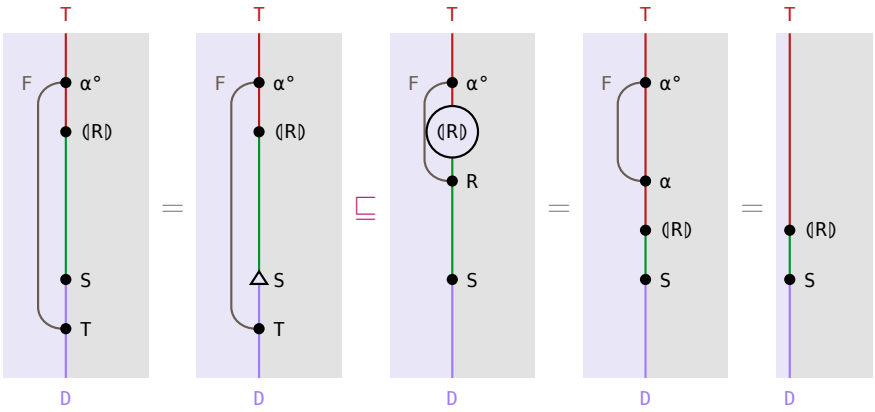
`(μX : φ(X))` is the least `X : A↛B` with `φ(X)⊑X`.

(6.2b)

	the law	what it says
1	`φ(Y) ⊑ Y ⇒ (μX : φ(X)) ⊑ Y`	to bound `(μX : φ(X))` above, exhibit one `Y` the body does not grow past — the half §Section 6.3. and every chapter after it uses
2	`Monotonic(φ) ⇒ φ(μX : φ(X)) = (μX : φ(X))`	**Knaster–Tarski**: the least solution of `φ(X)⊑X` already solves `φ(X)=X`, so the least prefix point and the least fixed point are one relation
3	`α° F(X) R ⊑ X ⇒ (|R|) ⊑ X` (6.2)	a relation `X` that contains one unfolding of the fold's recursion at `X` contains the fold
4	`X ⊑ α° F(X) R ⇒ X ⊑ (|R|)` (6.3)	a relation `X` contained in one unfolding of the fold's recursion at `X` is contained in the fold

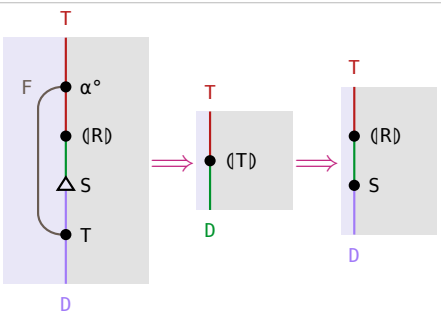
`F(S) T ⊑ R S ⇒ (|T|) ⊑ (|R|) S` (6.2c)

if **S** followed by **R** absorbs **T** after **F(S)**, the fold of **T** is below the fold of **R** followed by **S**



(1) (2) (3) (4) (5)

1	`α° F((|R|) S) T`	
2	`= α° F((|R|)) F(S) T`	`F(S) T ⊑ R S`
3	`⊑ α° F((|R|)) R S`	`α (|R|) = F((|R|)) R`
4	`= α° α (|R|) S`	`α° α = 𝟙`
5	`= (|R|) S`	

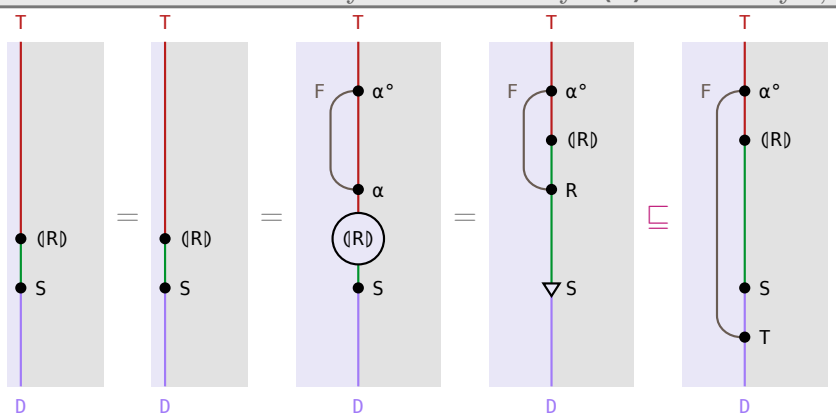


(1) (2) (3)

1	`F(S) T ⊑ R S`	`F(S) T ⊑ R S` `⇒ α° F((|R|) S) T ⊑ (|R|) S`
2	`⟹ α° F((|R|) S) T ⊑ (|R|) S`	`α° F(X) R ⊑ X` `⇒ (|R|) ⊑ X`
3	`⟹ (|T|) ⊑ (|R|) S`	

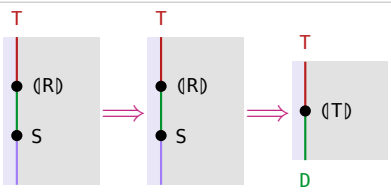
R S ⊑ F(S) T ⇒ (R) S ⊑ (T) (6.2d)

if **R** followed by **S** is absorbed by **F(S)** followed by **T**, the fold of **R** followed by **S** is below the fold of **T**



(1) (2) (3) (4) (5)

1	(R) S	
2	= (R) S	α° α = 𝟙
3	= α° α (R) S	α (R) = F((R)) R
4	= α° F((R)) R S	R S ⊑ F(S) T
5	⊑ α° F((R) S) T	



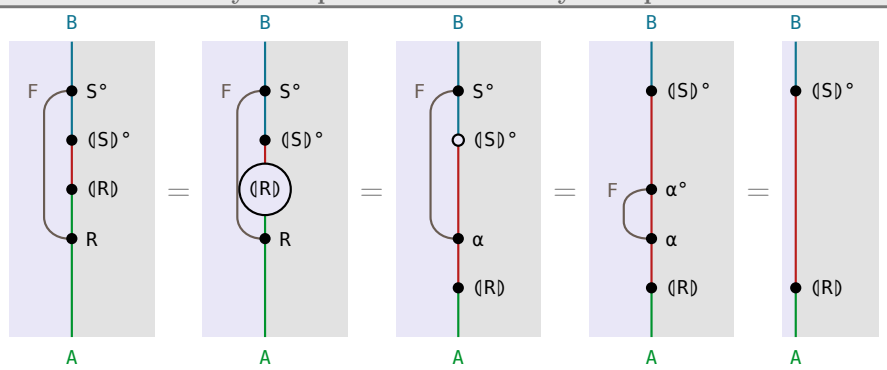
(1) (2) (3)

1	R S ⊑ F(S) T	R S ⊑ F(S) T ⇒ (R) S ⊑ α° F((R) S) T
2	⟹ (R) S ⊑ α° F((R) S) T	X ⊑ α° F(X) R ⇒ X ⊑ (R)
3	⟹ (R) S ⊑ (T)	

§6.3. Hylomorphisms

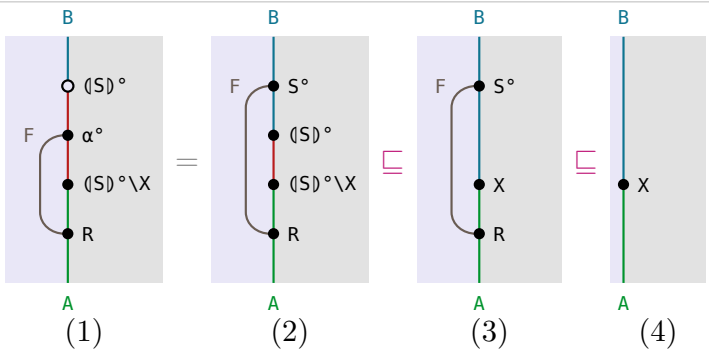
⦇S⦈° ⦇R⦈ = (μX : S° F(X) R) (6.3a)

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation



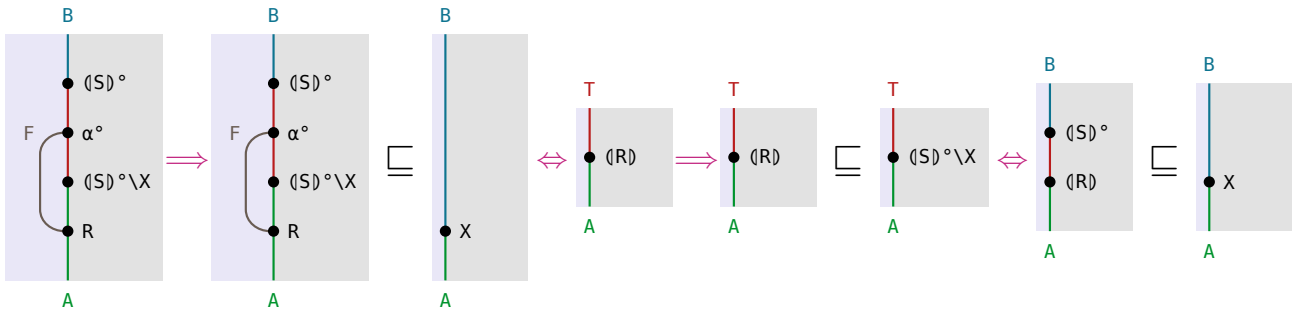
(1) (2) (3) (4) (5)

1	S° F(⦇S⦈° ⦇R⦈) R	
2	= S° F(⦇S⦈°) F(⦇R⦈) R	α ⦇R⦈ = F(⦇R⦈) R
3	= S° F(⦇S⦈°) α ⦇R⦈	⦇R⦈° α° = R° F(⦇R⦈°)
4	= ⦇S⦈° α° α ⦇R⦈	α° α = 𝟙
5	= ⦇S⦈° ⦇R⦈	



(1) (2) (3) (4)

1	⦇S⦈° α° F(⦇S⦈° \ X) R	⦇R⦈° α° = R° F(⦇R⦈°)
2	= S° F(⦇S⦈° (⦇S⦈° \ X)) R	S (S \ R) ⊑ R
3	⊑ S° F(X) R	S° F(X) R ⊑ X
4	⊑ X	

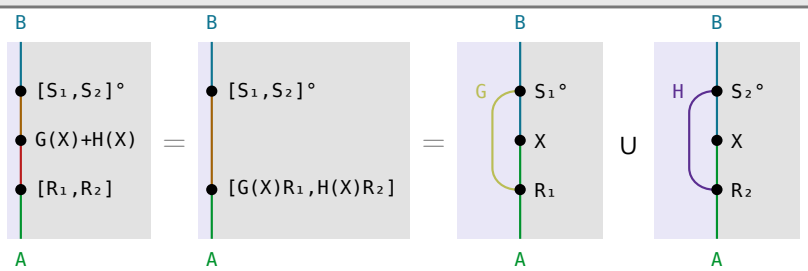


(1) (2) (3) (4) (5)

1	S° F(X) R ⊑ X	S° F(X) R ⊑ X ⇒ ⦇S⦈° α° F(⦇S⦈° \ X) R ⊑ X
2	⟹ ⦇S⦈° α° F(⦇S⦈° \ X) R ⊑ X	X ⊑ S \ R ⇔ S X ⊑ R
3	⟺ α° F(⦇S⦈° \ X) R ⊑ ⦇S⦈° \ X	α° F(X) R ⊑ X ⇒ ⦇R⦈ ⊑ X
4	⟹ ⦇R⦈ ⊑ ⦇S⦈° \ X	X ⊑ S \ R ⇔ S X ⊑ R
5	⟺ ⦇S⦈° ⦇R⦈ ⊑ X	

⦇[S₁,S₂]⦈° ⦇[R₁,R₂]⦈ = (μX : (S₁° G(X) R₁) ∪ (S₂° H(X) R₂)) (6.3b)

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results



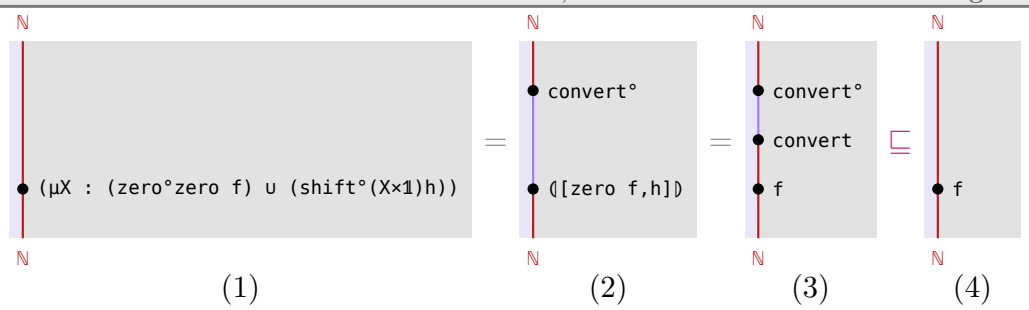
(1) (2) (3)

1	`[S₁,S₂]° (G(X) + H(X)) [R₁,R₂]`	`(R + S) [P,Q] = [R P,S Q]`
2	`= [S₁,S₂]° [G(X) R₁,H(X) R₂]`	`[U,V]° [R,S] = (U° R) ∪ (V° S)`
3	`= (S₁° G(X) R₁) ∪ (S₂° H(X) R₂)`	

§6.4. Fast exponentiation and modulus computation

shift f = (f × 𝟙) h ⇒ (μX : (zero° zero f) ∪ (shift° (X × 𝟙) h)) ⊑ f (6.4a)

once **f** turns **shift** into **h**, the recursion that halves the argument at each step computes **f**



(1) (2) (3) (4)

1	(μX : (zero° zero f) ∪ (shift° (X × 𝟙) h))	convert° ⦇[g,h]⦈ = (μX : (zero° g) ∪ (shift° (X × 𝟙) h))
2	= convert° ⦇[zero f,h]⦈	zero f = g ∧ shift f = (f × 𝟙) h ⇒ convert f = ⦇[g,h]⦈
3	= convert° convert f	Simple(convert)
4	⊑ f	

(μX : (zero° one) ∪ (shift° (X × 𝟙) op(a))) ⊑ exp(a) (6.4b)

the recursion that halves the exponent at each step computes **a** to the power **b**

`zero exp(a) = one`

`a` to the power `0` is `1`

`shift exp(a) = (exp(a) × 𝟙) op(a)`

`a` to the power `2n+d` is `op(a)` of `a` to the power `n` and `d`

(μX : (zero° zero) ∪ (shift° (X × 𝟙) op(b))) ⊑ mod(b) (6.4c)

the recursion that halves **a** at each step computes the remainder of **a** by **b**

`zero mod(b) = zero`

`0 mod b` is `0`

`shift mod(b) = (mod(b) × 𝟙) op(b)`

`(2a+d) mod b` is `op(b)` of `a mod b` and `d`

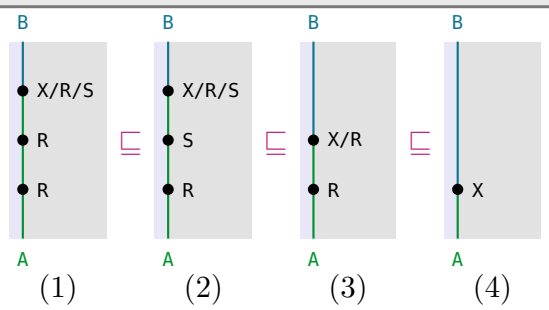
§6.5. Unique fixed points

$$R^2 \sqsubseteq S\ R \Rightarrow (X\ /\ R\ /\ S)\ R^2 \sqsubseteq X \tag{6.5a}$$

((X/R)/S) followed by RR stays below X when RR⊑SR

$$\mathsf{inductive}(S) \wedge R^2 \sqsubseteq S\ R \Rightarrow \mathsf{inductive}(R)$$

R is inductive when RR⊑SR for an inductive S



1	(X / R / S) R²	R² ⊑ S R
2	⊑ (X / R / S) S R	(R / S) S ⊑ R
3	⊑ (X / R) R	(R / S) S ⊑ R
4	⊑ X	

§6.5.1. Membership

$$\mathsf{member} = \mathbb{0} \tag{6.5.1a}$$

a constant relator records no elements, so its membership is empty

$$\mathsf{member} = [\mathsf{member}, \mathsf{member}]$$

a member of F+G is a member of whichever summand the value is in

$$\mathsf{member} = (\pi_1\ \mathsf{member}) \cup (\pi_2\ \mathsf{member})$$

a member of F×G is a member of either component

§6.5.2. Consequences

$$\mathsf{inductive}(S\ \mathsf{member}) \wedge X = S\ F(X)\ R \wedge Y = S\ F(Y)\ R \Rightarrow X = Y \tag{6.5.2a}$$

when S member is inductive, two solutions X, Y of X=SF(X)R are equal

$$\mathsf{inductive}(S\ \mathsf{member}) \wedge \mathsf{Entire}(S) \wedge \mathsf{Entire}(R) \wedge S\ F(X)\ R \sqsubseteq X \Rightarrow \mathsf{Entire}(X) \tag{6.5.2b}$$

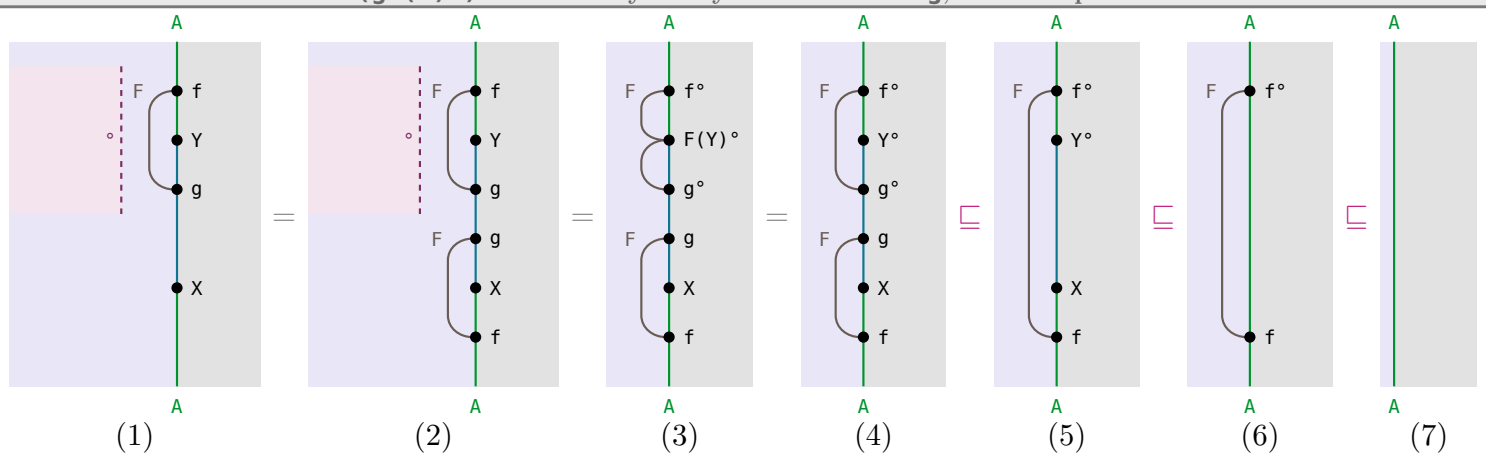
when S member is inductive and S, R are entire, a pre-fixed point of X↦SF(X)R is entire

$$\mathsf{inductive}(g\ \mathsf{member}) \wedge \mathsf{Map}(g) \wedge \mathsf{Map}(f) \wedge X = g\ F(X)\ f \Rightarrow \mathsf{Map}(X) \tag{6.5.2c}$$

when g member is inductive and f, g are maps, the solution of X=gF(X)f is a map

$$X = g\ F(X)\ f \wedge \mathsf{Simple}(g) \wedge \mathsf{Simple}(f) \wedge Y^\circ\ X \sqsubseteq \mathbb{1} \Rightarrow (g\ F(Y)\ f)^\circ\ X \sqsubseteq \mathbb{1}$$

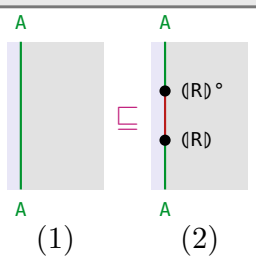
(gF(Y)f)° followed by X stays below 𝟙 when g, f are simple and Y°X⊑𝟙



1	(g F(Y) f)° X	X = g F(X) f
2	= (g F(Y) f)° g F(X) f	(R S)° = S° R°
3	= f° F(Y)° g° g F(X) f	PreservesRecip(F)
4	= f° F(Y°) g° g F(X) f	Simple(g)
5	⊑ f° F(Y° X) f	Y° X ⊑ 𝟙
6	⊑ f° f	Simple(f)
7	⊑ 𝟙	

inductive(R° member) ∧ 𝟙 ⊑ R° R ⇒ 𝟙 ⊑ (R)° (R) (6.5.2d)

when **R°member** is inductive and **R** is surjective, the fold **(R)** is surjective

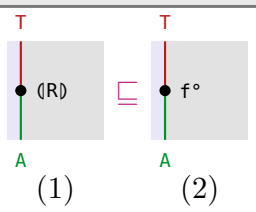


(1) (2)

1	𝟙	inductive(S member) ∧ Entire(S) ∧ Entire(R) ∧ S F(X) R ⊑ X ⇒ Entire(X) at S=R°
2	⊑ (R)° (R)	

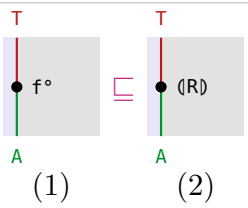
Map(f) ∧ R f ⊑ F(f) α ∧ 𝟙 ⊑ R° R ⇒ f° = (R) (6.5.2e)

a map **f** with **Rf⊑F(f)α**, for a surjective **R**, has the fold **(R)** as its converse



(1) (2)

1	(R)	shunting f, fusion, Rf⊑F(f)α
2	⊑ f°	

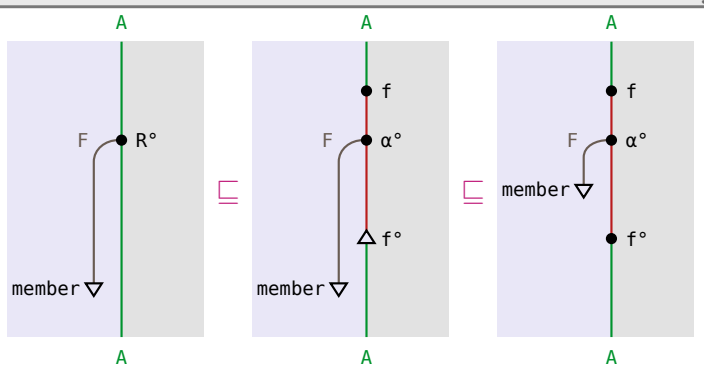


(1) (2)

1	f°	inductive(R° member) ∧ 𝟙 ⊑ R° R ⇒ 𝟙 ⊑ (R)° (R) and (R)f⊑𝟙
2	⊑ (R)	

Map(f) ∧ R f ⊑ F(f) α ⇒ R° member ⊑ f α° member f° (6.5.2f)

R°member is below **α°member** conjugated by **f**, so it is inductive when **α°member** is



(1) (2) (3)

1	R° member	Map(f) ∧ R f ⊑ F(f) α ⇒ R° ⊑ f α° F(f°)
2	⊑ f α° F(f°) member	F(R) member ⊑ member R
3	⊑ f α° member f°	

§6.6. Sorting by selection

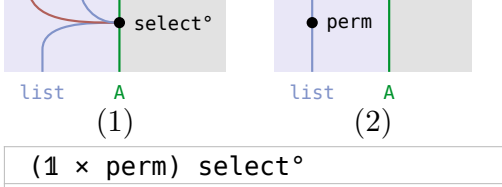
§6.6.1. The function sort

(6.6.1a)

	definition	type	meaning
1	`ordered[] = true, ordered([a]#xs) = (∀x. x ∈ xs ⇒ R(a,x)) ∧ ordered(xs)`	`[A]→ [A]`	every element is R-below each element after it
2	`(a,xs) ok q ⇔ (a,xs) = q ∧ (∀x. x ∈ xs ⇒ R(a,x))`	`A×[A]→ A×[A]`	the coreflexive passing `(a, x)` when `a` is R-below every element of `x`

`ordered = (|[nil,ok cons]|)` (6.6.1b)

a list is sorted exactly when the fold that rebuilds it passes **`ok`** at every **`cons`**, i.e. each head is **R**-below every element after it



1	`ordered`	fold uniqueness
2	`= (|[nil,ok cons]|)`	

§6.6.2. Selection sort

`select° ⊑ ok cons perm ⇒ (|[nil,select°]|)° ⊑ perm ordered` (6.6.2a)

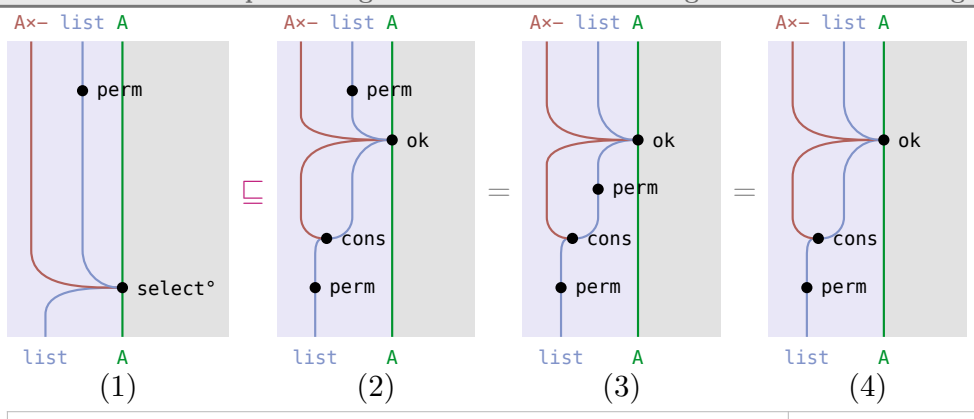
every output of unfolding the input by **`select`** is a sorted permutation of it



1	`(|[nil,select°]|)°`	`select° ⊑ ok cons perm` `⇒ (|[nil,select°]|)° ⊑ ((|[nil,ok cons]|) perm)°`
2	`⊑ ((|[nil,ok cons]|) perm)°`	(6.6.1b) `ordered = (|[nil,ok cons]|)`
3	`= (ordered perm)°`	`(R S)° = S° R°`
4	`= perm° ordered°`	`perm° = perm`
5	`= perm ordered°`	`ordered° = ordered`
6	`= perm ordered`	

`select° ⊑ ok cons perm ⇒ (𝟙 × perm) select° ⊑ ok cons perm` (6.6.2b)

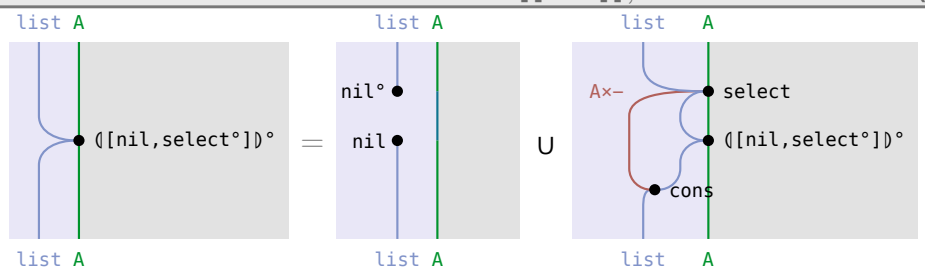
permuting the tail and then undoing **`select`** lands among the **`ok`** conses of a permutation



1	`(𝟙 × perm) select°`	`select° ⊑ ok cons perm`
2	`⊑ (𝟙 × perm) ok cons perm`	`ok (𝟙 × perm) = (𝟙 × perm) ok`
3	`= ok (𝟙 × perm) cons perm`	`cons perm = (𝟙 × perm) cons perm`
4	`= ok cons perm`	

`◖[nil,select°]◗° = (nil° nil) ∪ (select (𝟙 × ◖[nil,select°]◗°) cons)` (6.6.2c)

selection sort returns `[]` on `[]`, and otherwise selects `(a,y)`, sorts `y` and conses `a` back on

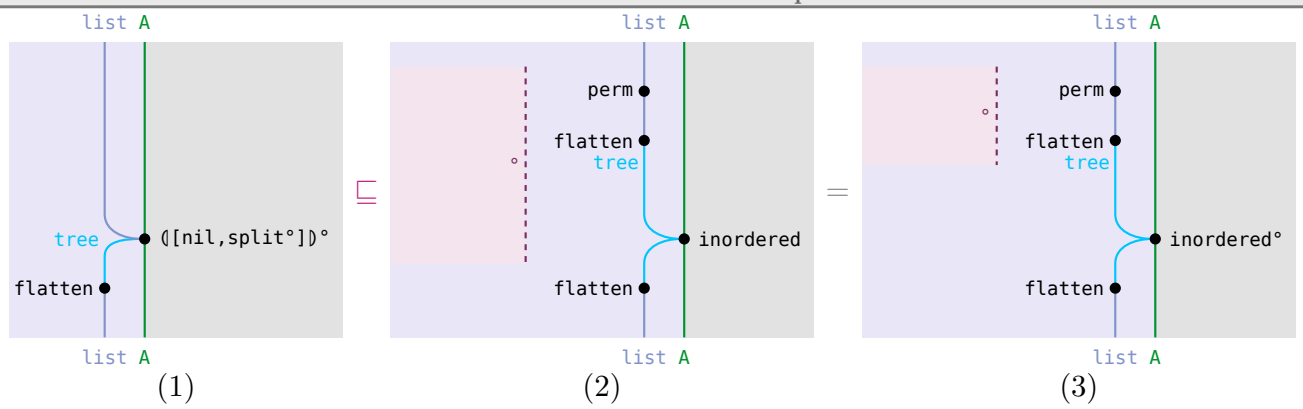


1	`◖[nil,select°]◗°`	unfolding the converse of a fold
2	`= (nil° nil) ∪ (select (𝟙 × ◖[nil,select°]◗°) cons)`	

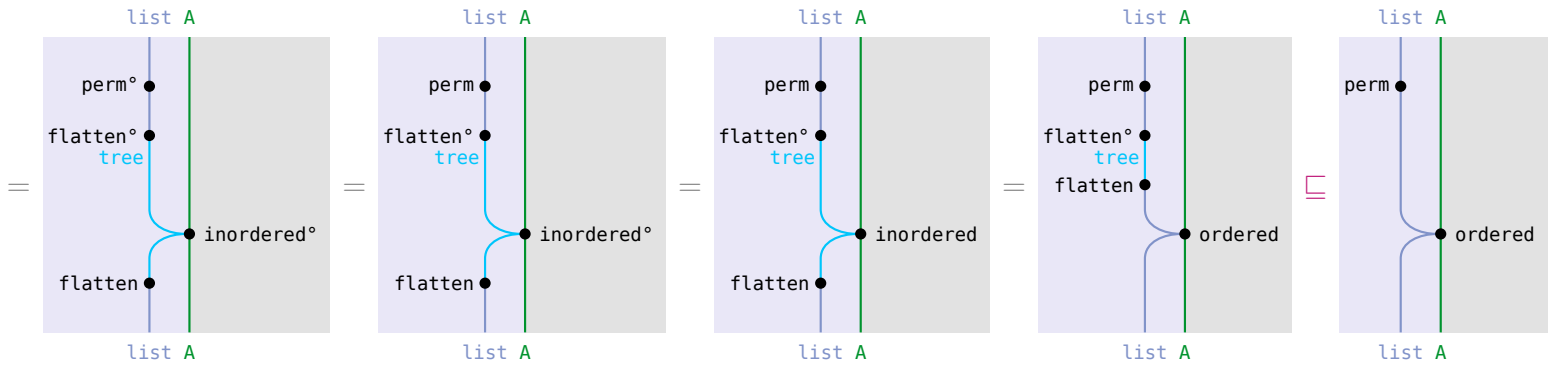
§6.6.3. Quicksort

`(∀a. ∀b. ∀c. R(a,b) ⇒ R(b,c) ⇒ R(a,c)) ∧ split° ⊑ check' join perm ⇒ ◖[nil,split°]◗° flatten ⊑ perm ordered` (6.6.3a)

every output of unfolding the input by `split` into a tree and flattening that tree is a sorted permutation of the input



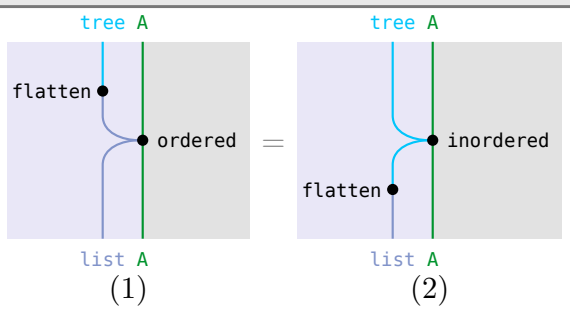
1	`◖[nil,split°]◗° flatten`	`split° ⊑ check' join perm ⇒ ◖[nil,split°]◗° flatten ⊑ (inordered flatten perm)° flatten`
2	`⊑ (inordered flatten perm)° flatten`	`(R S)° = S° R°`
3	`= (flatten perm)° inordered° flatten`	`(R S)° = S° R°`



4	`= perm° flatten° inordered° flatten`	`perm° = perm`
5	`= perm flatten° inordered° flatten`	`inordered° = inordered`
6	`= perm flatten° inordered flatten`	(6.6.3b) `(∀a. ∀b. ∀c. R(a,b) ⇒ R(b,c) ⇒ R(a,c)) ⇒ flatten ordered = inordered flatten`
7	`= perm flatten° flatten ordered`	`flatten° flatten ⊑ 𝟙`
8	`⊑ perm ordered`	

(∀a. ∀b. ∀c. R(a,b) ⇒ R(b,c) ⇒ R(a,c)) ⇒ flatten ordered = inordered flatten (6.6.3b)

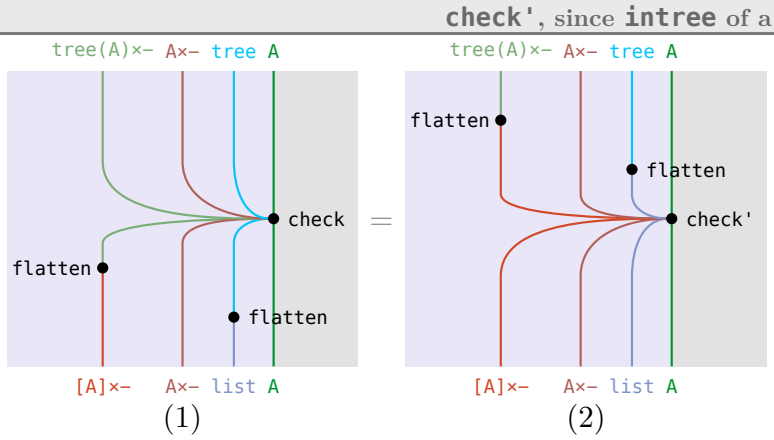
flattening a tree and then testing the list for order is the same as testing the tree with **inordered** and then flattening



1	flatten ordered	R transitive: a flattened tree is ordered iff every node passes check
2	= inordered flatten	

check (flatten × (𝟙 × flatten)) = (flatten × (𝟙 × flatten)) check' (6.6.3c)

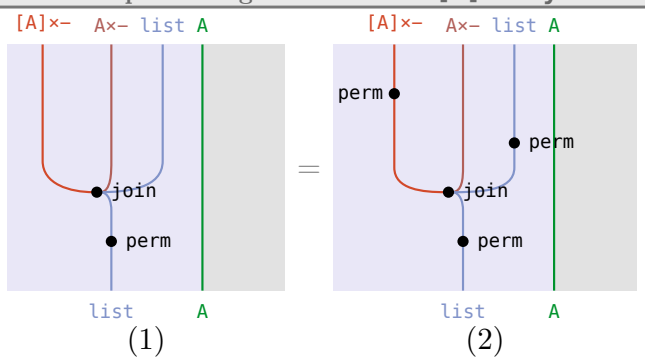
testing a fork with **check** and then flattening both subtrees equals flattening both subtrees and then testing with **check'**, since **intree** of a tree is **inlist** of its flattening



1	check (flatten × (𝟙 × flatten))	b intree x ⇔ b inlist flatten(x)
2	= (flatten × (𝟙 × flatten)) check'	

join perm = (perm × (𝟙 × perm)) join perm (6.6.3d)

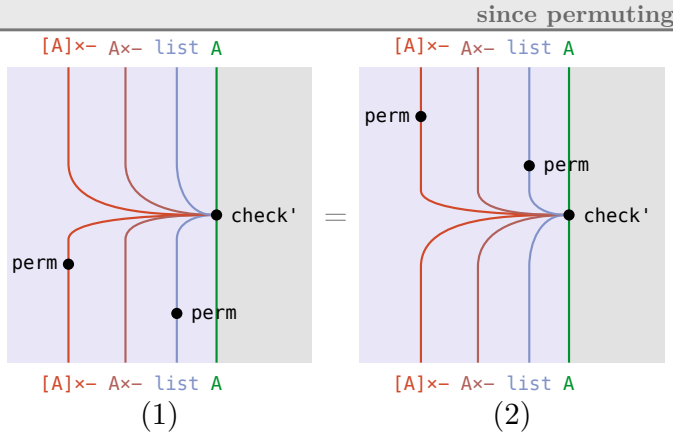
permuting the list **x ++ [a] ++ y** is the same as first permuting **x** and **y** and then permuting the joined list



1	join perm	perm is transitive and closed under ++
2	= (perm × (𝟙 × perm)) join perm	

check' (perm × (𝟙 × perm)) = (perm × (𝟙 × perm)) check' (6.6.3e)

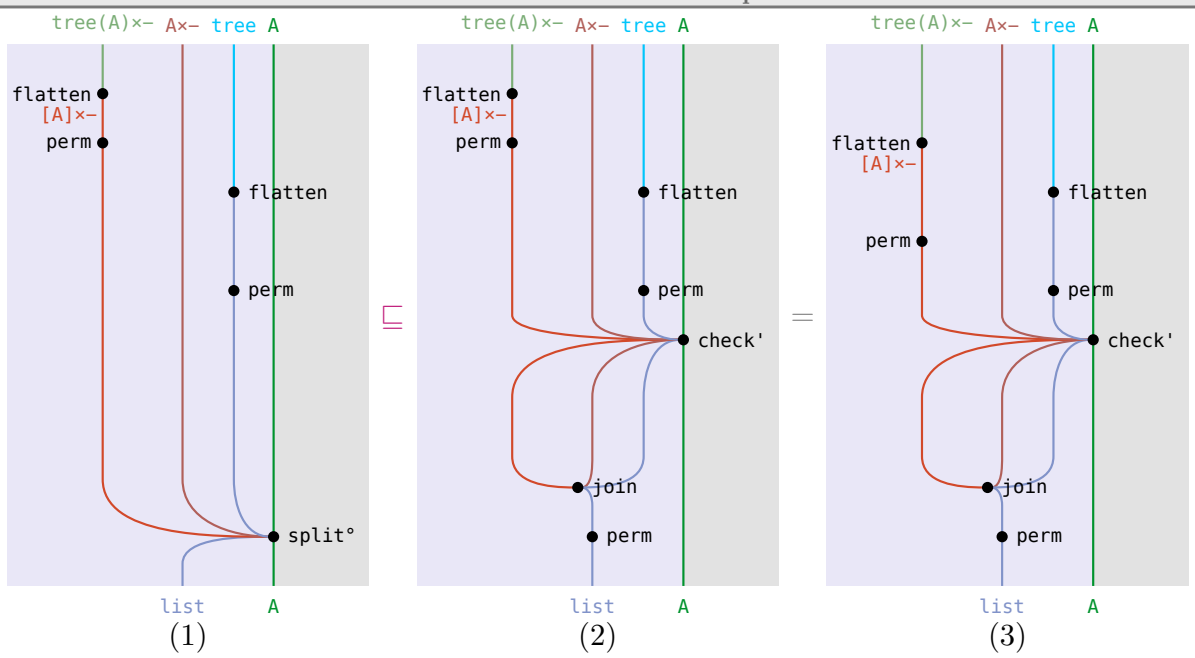
testing a fork of lists with **check'** and then permuting both parts equals permuting both parts and then testing, since permuting a list leaves its members unchanged



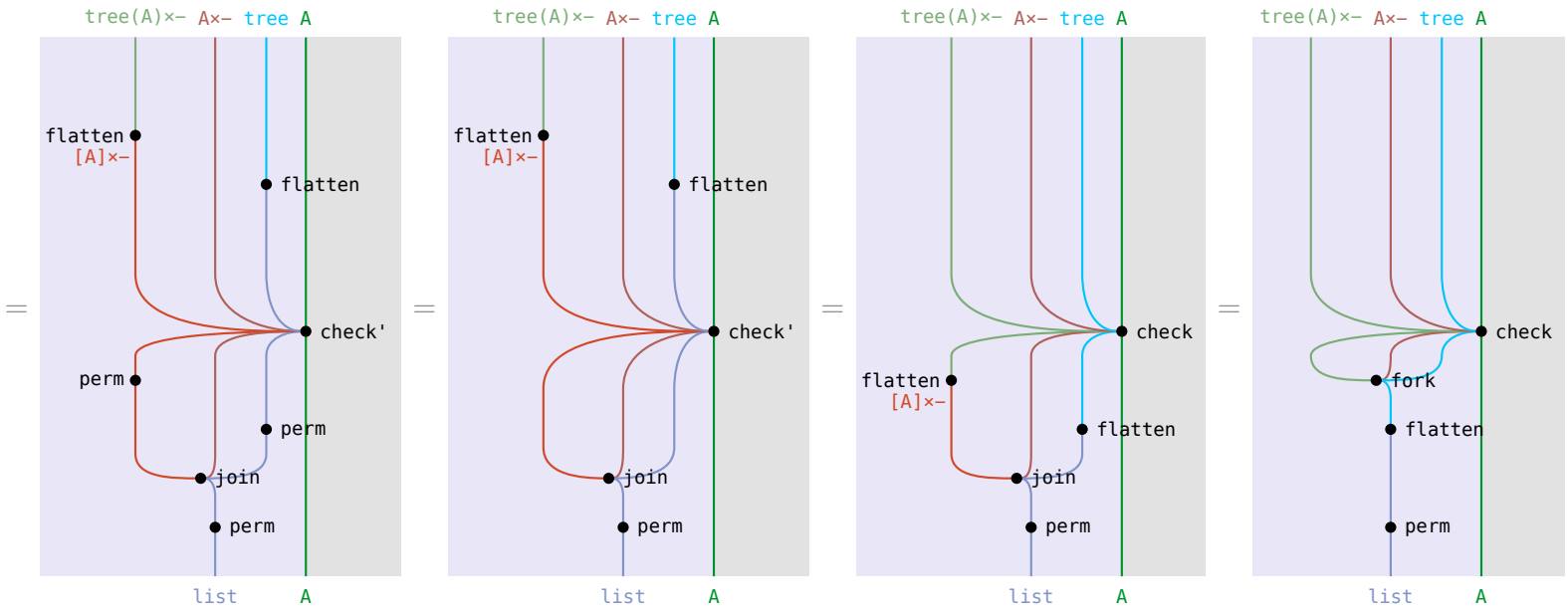
1	check' (perm × (𝟙 × perm))	b inlist x ⇔ b inlist perm(x)
2	= (perm × (𝟙 × perm)) check'	

split° ⊑ check' join perm ⇒ ((flatten perm) × (𝟙 × (flatten perm))) split° ⊑ check fork flatten perm (6.6.3f)

undoing **split** and then flattening and permuting both parts lands among the **check**ed forks whose flattening is permuted



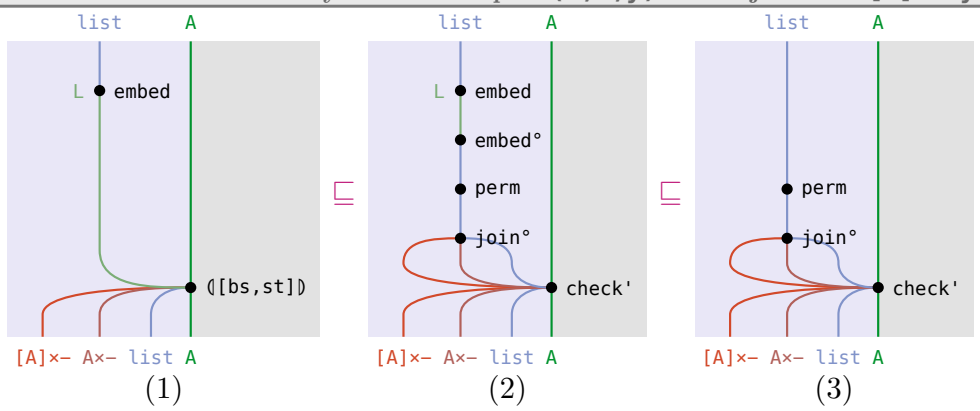
1	((flatten perm) × (𝟙 × (flatten perm))) split°	split° ⊑ check' join perm
2	⊑ ((flatten perm) × (𝟙 × (flatten perm))) check' join perm	
3	= (flatten × (𝟙 × flatten)) (perm × (𝟙 × perm)) check' join perm	(6.6.3e) check' (perm × (𝟙 × perm)) = (perm × (𝟙 × perm)) check'



4	= (flatten × (𝟙 × flatten)) check' (perm × (𝟙 × perm)) join perm	(6.6.3d) join perm = (perm × (𝟙 × perm)) join perm
5	= (flatten × (𝟙 × flatten)) check' join perm	(6.6.3c) check (flatten × (𝟙 × flatten)) = (flatten × (𝟙 × flatten)) check'
6	= check (flatten × (𝟙 × flatten)) join perm	fork flatten = (flatten × (𝟙 × flatten)) join
7	= check fork flatten perm	

bs ⊑ τ perm join° check' ∧ (𝟙 × (perm join° check')) st ⊑ cons perm join° check' ⇒ embed ⦇[bs,st]⦈ ⊑ perm join° check' (6.6.3g)

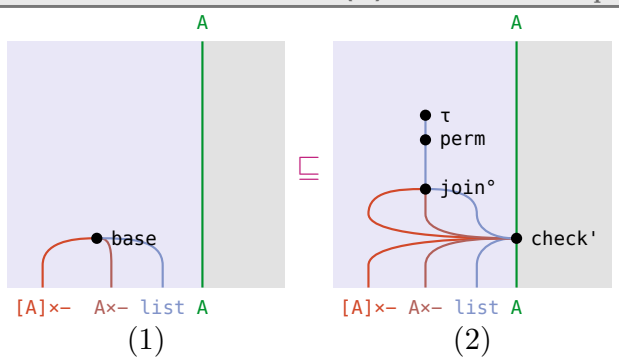
turning a list into a non-empty list and folding it with any **[bs, st]** meeting the two conditions below gives only **check**ed triples **(x,a,y)** whose join **x ⧺ [a] ⧺ y** is a permutation of the list



1	embed ⦇[bs,st]⦈	bs ⊑ τ perm join° check' ∧ (𝟙 × (perm join° check')) st ⊑ cons perm join° check' ⇒ embed ⦇[bs,st]⦈ ⊑ embed embed° perm join° check'
2	⊑ embed embed° perm join° check'	embed embed° perm join° check' ⊑ perm join° check'
3	⊑ perm join° check'	

base ⊑ τ perm join° check' (6.6.3h)

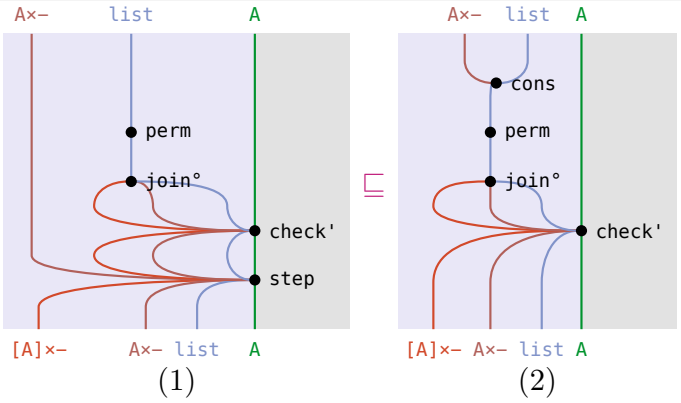
base(a) is a **check**ed triple whose join is a permutation of the one-element list **[a]**



1	base	[] ⧺ [a] ⧺ [] = [a], and check' holds on empty lists
2	⊑ τ perm join° check'	

connected(R) ⇒ (𝟙 × (perm join° check')) step ⊑ cons perm join° check' (6.6.3i)

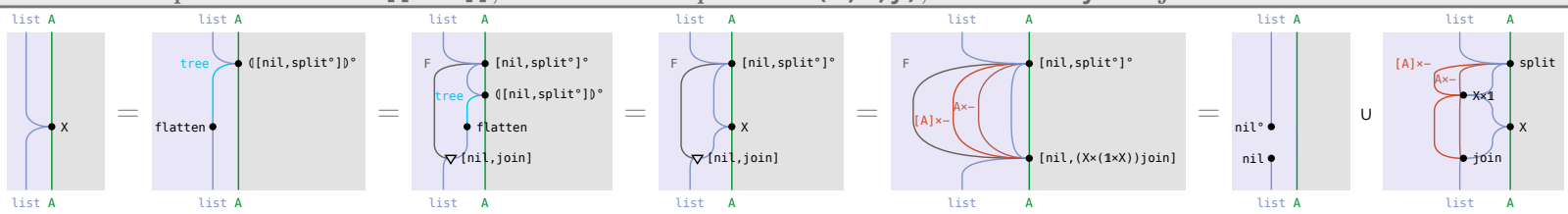
if **(x,b,y)** is a **check**ed triple for a permutation of **l**, then **step(a,(x,b,y))** is a **check**ed triple for a permutation of **[a] ⧺ l**



1	(𝟙 × (perm join° check')) step	aRb puts a in front of x; otherwise bRa and a goes in front of y
2	⊑ cons perm join° check'	

X = ⦇[nil,split°]⦈° flatten ⇒ X = (nil° nil) ∪ (split (X × (𝟙 × X)) join) (6.6.3j)

quicksort returns [] on [], and otherwise splits into (x,a,y), sorts x and y and joins them around a

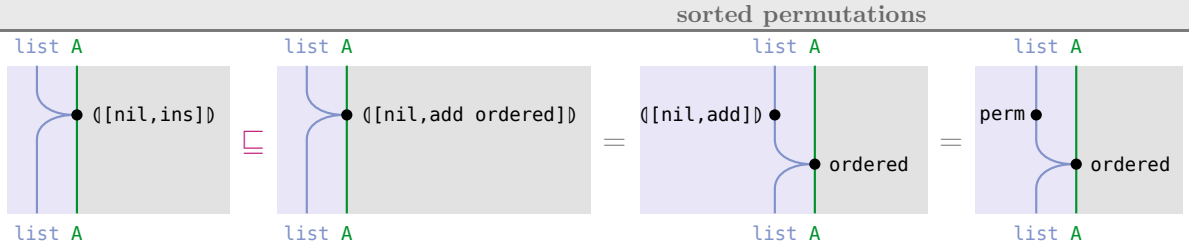


(1) (2) (3) (4) (5) (6)

1	X	X = ⦇[nil,split°]⦈° flatten
2	= ⦇[nil,split°]⦈° flatten	S° F(⦇S⦈° ⦇R⦈) R = ⦇S⦈° ⦇R⦈
3	= [nil,split°]° F(⦇[nil,split°]⦈° flatten) [nil,join]	X = ⦇[nil,split°]⦈° flatten
4	= [nil,split°]° F(X) [nil,join]	F(S) [T,U] = [T,(S × (𝟙 × S)) U]
5	= [nil,split°]° [nil,(X × (𝟙 × X)) join]	[U,V]° [R,S] = (U° R) ∪ (V° S)
6	= (nil° nil) ∪ (split (X × (𝟙 × X)) join)	

(𝟙 × ordered) ins ⊑ add ordered ⇒ ⦇[nil,ins]⦈ ⊑ perm ordered (6.6.3k)

for any ins that, on an ordered list, returns only ordered results of add, folding with [nil, ins] gives only sorted permutations

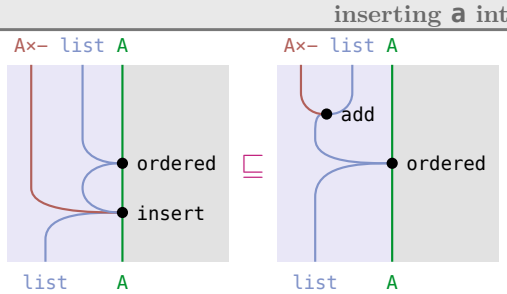


(1) (2) (3) (4)

1	⦇[nil,ins]⦈	(𝟙 × ordered) ins ⊑ add ordered ⇒ ⦇[nil,ins]⦈ ⊑ ⦇[nil,add ordered]⦈
2	⊑ ⦇[nil,add ordered]⦈	⦇[nil,add]⦈ ordered = ⦇[nil,add ordered]⦈
3	= ⦇[nil,add]⦈ ordered	perm = ⦇[nil,add]⦈
4	= perm ordered	

connected(R) ∧ (∀a. ∀b. ∀c. R(a,b) ⇒ R(b,c) ⇒ R(a,c)) ⇒ (𝟙 × ordered) insert ⊑ add ordered (6.6.3l)

inserting a into an ordered list puts a somewhere in it and keeps it ordered



(1) (2)

1	(𝟙 × ordered) insert	insert splices a in, and R transitive and connected
2	⊑ add ordered	

§6.7. Closure

$$R^* = (\mu X : \mathbb{1} \cup (X\ R)) \tag{6.7a}$$

closing R by composing it on the left and closing it by composing on the right give the same relation R^*

$$\mathbb{1} \sqsubseteq X \wedge X^2 \sqsubseteq X \Rightarrow R \sqsubseteq X \Leftrightarrow R^* \sqsubseteq X \tag{6.7b}$$

X contains R exactly when it contains R^*

$$\mathbb{1} \sqsubseteq R^* \tag{6.7c}$$

R^* is reflexive

1	$\mathbb{1}$	$R \sqsubseteq R \cup S$
2	$\sqsubseteq \mathbb{1} \cup (R\ R^*)$	$\mathbb{1} \cup (R\ R^*) = R^*$
3	$= R^*$	

$$R \sqsubseteq R^* \tag{6.7d}$$

R^* contains R

1	R	
2	$= R$	(6.7c) $\mathbb{1} \sqsubseteq R^*$
3	$\sqsubseteq R\ R^*$	$R\ R^* \sqsubseteq R^*$
4	$\sqsubseteq R^*$	

$$(\mathbb{1} \cup (R\ (R^* / R^*)))\ R^* \sqsubseteq R^* \tag{6.7e}$$

R^*/R^* is a prefixed point of $X \mapsto \mathbb{1} \cup RX$, stated through division

$$R^{*2} \sqsubseteq R^*$$

R^* is transitive, because R^*/R^* is a prefixed point of $X \mapsto \mathbb{1} \cup RX$

1	$(\mathbb{1} \cup (R\ (R^* / R^*)))\ R^*$	$(S \cup T)\ R = (S\ R) \cup (T\ R)$
2	$= (R^*) \cup (R\ (R^* / R^*)\ R^*)$	
3	$= R^* \cup (R\ (R^* / R^*)\ R^*)$	$(R / R)\ R \sqsubseteq R$
4	$\sqsubseteq R^* \cup (R\ R^*)$	$R\ R^* \sqsubseteq R^*$
5	$\sqsubseteq R^*$	

$$\mathbb{1} \sqsubseteq X \wedge X^2 \sqsubseteq X \wedge R \sqsubseteq X \Rightarrow \mathbb{1} \cup (R\ X) \sqsubseteq X \tag{6.7f}$$

a reflexive transitive X containing R is a prefixed point of $Y \mapsto \mathbb{1} \cup RY$

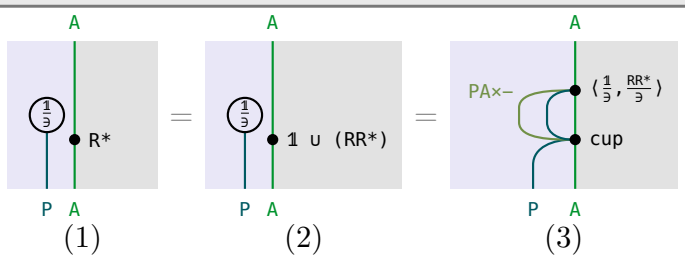
$$\mathbb{1} \sqsubseteq X \wedge X^2 \sqsubseteq X \wedge R \sqsubseteq X \Rightarrow R^* \sqsubseteq X$$

a reflexive transitive X containing R is a prefixed point of $X \mapsto \mathbb{1} \cup RX$, so it contains R^*

1	$\mathbb{1} \cup (R\ X)$	$\mathbb{1} \sqsubseteq X$
2	$\sqsubseteq X \cup (R\ X)$	$R \sqsubseteq X$
3	$\sqsubseteq X \cup (X^2)$	
4	$\sqsubseteq X$	

$$\frac{R^*}{\ni} = \langle \frac{\mathbb{1}}{\ni}, \frac{R\ R^*}{\ni} \rangle\ \mathrm{cup} \tag{6.7g}$$

the set of R^*-successors of a is a itself joined with the R^*-successors of its R-successors



1	$\frac{R^*}{\ni}$	$\mathbb{1} \cup (R\ R^*) = R^*$
2	$= \frac{\mathbb{1} \cup (R\ R^*)}{\ni}$	$\frac{R \cup S}{\ni} = \langle \frac{R}{\ni}, \frac{S}{\ni} \rangle\ \mathrm{cup}$
3	$= \langle \frac{\mathbb{1}}{\ni}, \frac{R\ R^*}{\ni} \rangle\ \mathrm{cup}$	

§6.7.1. Computing closure

R − 𝟘 = R (6.7.1a)

taking nothing away leaves R

R ∪ (S − R) = R ∪ S

beside R, only the part of S outside R adds anything

R − (S ∪ T) = R − S − T

taking away S and then T takes away S∪T

(R ∪ S) − T = (R − T) ∪ (S − T)

taking T away from a union takes it away from each part

monotonic(φ) ∧ monotonic(ψ) ⇒ (μX : φ(ψ(X))) = φ(μY : ψ(φ(Y))) (6.7.1b)

the least fixed point of φ after ψ is φ applied to the least fixed point of ψ after φ

S R* = (μX : S ∪ (X R)) (6.7.1c)

S followed by any number of R steps is the least X containing S and closed under a further R step

R* S = (μX : S ∪ (R X))

any number of R steps followed by S is the least X containing S and closed under an R step in front

θ(𝟘,S) = S R* (6.7.1d)

started with nothing found and S to explore, θ computes SR*

1	θ(𝟘,S)	θ(P,Q) ≜ P ∪ (μX : Q ∪ ((X R) − P))
2	≜ 𝟘 ∪ (μX : S ∪ ((X R) − 𝟘))	(6.7.1a) R − 𝟘 = R
3	= 𝟘 ∪ (μX : S ∪ (X R))	𝟘 ∪ R = R
4	= (μX : S ∪ (X R))	(6.7.1c) S R* = (μX : S ∪ (X R))
5	= S R*	

θ(P,𝟘) = P (6.7.1e)

with nothing left to explore, θ returns what it has found

1	θ(P,𝟘)	θ(P,Q) ≜ P ∪ (μX : Q ∪ ((X R) − P))
2	≜ P ∪ (μX : 𝟘 ∪ ((X R) − P))	(μX : 𝟘 ∪ ((X R) − P)) = 𝟘
3	= P ∪ 𝟘	R ∪ 𝟘 = R
4	= P	